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RE: Reference | Yahya Jabary

To whom it may concern:

It is our pleasure to recommend Yahya Jabary. I and my team have known him since late 2025, when Yahya Jabary started to collaborate with our research group on assembly level optimizations using our xDSL compiler infrastructure. Starting with no experience in compiler development, he independently prototyped a lowering pipeline from LLVM IR to Apple AIR targeting MetalLib, implemented in Python. This required reverse engineering intermediate representations in Apple's GPU toolchain, for which no public specification exists.

Since the beginning of the year, we have been collaborating with Yahya on extending the code generation functionality in xDSL, our research group's largest project. In that time, Yahya has contributed the most commits of any external contributor (86). We have a rigorous review process, and Yahya has been fast to address feedback, and to incorporate our development style into his contributions. To streamline the process of new contributors, Yahya has established a document detailing xDSL's contributing guidelines. Recently, he has become a maintainer of the project, giving thoughtful and detailed reviews. The code generation functionality that Yahya has built has enabled prototypes for future research projects planned in our group. To stress-test this infrastructure, Yahya has implemented a transformer model based on Karpathy's MicroGPT using hand-written Exo kernels, covering attention, layer normalization, embeddings, and full transformer blocks. The resulting forward pass achieves 35 microseconds mean latency, compared to 185 microseconds for PyTorch and 503 microseconds for JAX under identical conditions. We are grateful for Yahya's contributions to our project and look forward to leveraging his work in our group's research on code generation using MLIR and xDSL.

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Out of the students that have reached out to work on xDSL over the last year, Yahya stands out as the most productive and pleasant to work with. Yahya is enthusiastic, proactive, and not afraid to tackle tasks that are exploratory in nature. He surfaces blockers early, and documents his thought process clearly. When introduced to new domains or tools with a large surface area, such as xDSL, Exo, and llvmlite, Yahya approached them thoughtfully, planning out the work required to use them well. Much of the work was self-directed, requiring little supervision time from us.

Yahya's results and progress over the few months of work together indicate that he is an excellent hire in both academia and industry.

Sincerely,

Tobias Grosser (Supervising Professor)

Alexandre Lopoukhine (Supervising PhD student)

Short Biography: Tobias Grosser

I am an Associate Professor at the University of Cambridge, researching compiler design for high-performance computing as well as emerging domains such as AI, crypto, and hardware design, and have been heavily invested in formal methods for compilation for the past 5 years. I obtained my PhD as a Google PhD Fellow at ENS Paris/UPMC, worked as an SNSF Ambizione Fellow at ETH Zurich, and served as a Reader at the University of Edinburgh. I have been an open-source compiler developer for over a decade and am a top-2% contributor to the LLVM compilation stack (by commit count). I co-founded open-source projects such as LLVM's Polly, CIRCT/LLHD, xDSL, and the polyhedral library FPL; I also regularly organize compiler events, such as the early Euro LLVM conferences or the MLIR (Un)School; and co-organized CGO 2024/2025.